

Interconnected ISPs

Laptop/phone/servers/IoT Device

**Internet:** Network of networks Billions of connected devices (hosts/end systems) using routers/any medium

Physical Media

Fibre Optics  
Copper wire  
Radio Waves

Packet Switches

Routers  
Switches

Internet Provides Services To Applications:

web Browsing, video streaming, online Gaming, Email, Cloud Services

**Protocol:** Format & order of messages Action taken when sending/receiving  
↳ Controls communication b/w computers (HTTP, TCP, IP, Wifi, Ethernet)

Protocols are Standardized by RFC (req. for comments) & IETF (Internet Engineering Task Force)

**Network Structure:**

→ Network Edge Clients, Server (often in Data Centre)

→ Access Network → ISP with DSL/Fibre/Wifi

→ Network Core Interconnected Routers Moves packet

Enterprise Model

- Companies/universities
- mix of wire/wireless links
- connecting mix of switches/routers

**Access Network:** i) Cable Based

HFC (Hybrid Fiber Coax)

Shared Networks

Uses FDM

High downstream speed  
40 MBps - 1.2 Gbps  
38-100 MBps

Shared Bandwidth

ii) DSL

uses existing telephone lines

Dedicated connection to central office

Voice & data separated by frequencies

24-52 Mbps

3.5 - 16 Mbps

iii) Wireless Access

• Wifi (local) WLAN

• 4G/5G (cellular)

wide Area cellular

• Connects end system to routers

**Data Centre Networks** High Bandwidth links (10s to 100s of Gbps), Connects 100s to 1000s of server together and to internet

**Host** Host sending func: take application message → breaks into smaller chunks called packets of length  $L$  bits. → Transmit packets into access network at trans rate  $R$

$$\text{Trans Delay} = \frac{\text{time needed to transmit } L \text{ bits}}{L} = \frac{L}{R}$$

**Links: Physical Media**

bit propagates b/w sender/receiver

physical link → lies b/w

guided media → physical media

Twisted Pair (Insulated Copper wires)

Coaxial → bidirectional

Fibre optic Hair thin, light Pulses, Low error

Unguided media → Nonphysical Media

Wireless Radio signals carried EM spectrum

Propagation env effects

Radio Link types

Half Duplex, Non Physical

unicast → To single  
Multicast → To multiple

Broadcast → general announcement

wireless LAN

Wide Area

Bluetooth

microwave satellite



# IP Addressing

Logical Addressing

IPv4 - 32bit dotted decimal Notation  $\rightarrow$  Normally used  
 IPv6 128bit Hexa colon notation

IP Address:

MAC/Physical Address: 48bit 1 hexa  $\rightarrow$  fixed can't be changed

IPv4: 32bits are divided into 4 octets known as dotted decimal Notation

10101100.00010000.00000001.01001110

172.16.1.78

128 64 32 16 8 4 2 1

1 0 1 0 1 1 0 0 = 172

1 1 1 1 1 1 1 1 = 255

## IP Address classes:

IP Addr Scheme is divided into 5 classes

IP address can't be greater than 255

Class A (0-126) 127 loop back IP

Class B 128-191

Class C 192-223

Class D 224-239

Class E 240-255

In Binary  $\rightarrow$  0

For class recognition 10

110

1110

1111

202.10.20.5  $\in$  Class C

### RANGES

A [ 0.0.0.0 to

127.255.255.255

0xxx  
127xxx reserved

128.0.0.0 to

191.255.255.255

192.0.0.0

223.255.255.255

224.0.0.0

239.255.255.255

240.0.0.0

255.255.255.255

IP

-2: Network Addr  
Broadcast Addr

Network Host

Hosts

101 A: N.H.H.H Fix  $10 \cdot x \cdot x \cdot x$   $2^{24} - 2$

110 B: N.N.H.H  $172 \cdot 16 \cdot x \cdot x$   $2^{16} - 2$

111 C: N.N.N.H  $192 \cdot 168 \cdot 1 \cdot x$   $2^8 - 2$

Fix

Network Addr:

220.10.5.0

Broadcast Addr:

220.10.5.255

All 1's or

255 in Host octet

$$A = 2^{8-1} = 2^7 = 128$$

$$128 - 2 = 126$$

$$B = 2^{16-2} = 2^{14} = 16384$$

$$C = 2^{24-3} = 2^{21}$$



**Private IP Addr:** Certain IPs are reserved for LAN these are Private IP Addr, used for Home, office Network. Not connected to Internet.

ex: 10.0.0.0 to 10.255.255.255

**Subnet Mask:** 172.16.1.78 AND = 172.16.0.0 → NW Addr  
 Differentiate Net mask → 255.255.0.0 for class B  
 & Host portion don't change this  
 All 1's in NW & All 0's in Host

VLSM  
VLSM

$2^n - 2 \Rightarrow \text{Req}$  host = 0 host bit 1 = broadcast  
 network = 1

/24

XX → Host Bits 32 - XX = H

/22

11111111.11111111.11111111.00000000

255.255.255.0

11111111.11111111.11111100.00000000

255.255.252.0

Subnetting

→ Break your network into smaller networks

Problems of Big Networks

Security  
 Broadcasting Domain  
 Collision Domain  
 Troubleshooting

192.168.1.6 /24 Network  
 192.168.1.1 /24 1st IP  
 192.168.1.254 /24 Last IP  
 192.168.1.255 /24 Broadcasting

Make Network two Subnetworks

192.168.1.00000000 /24 Host = 32 - 24 = 8

192.168.1.00000000 /25 H = 32 - 25 = 7

192.168.1.10000000 /25 H = 32 - 25 = 7

Network

Host

11 ⇒ 2<sup>11</sup> = 2048 → FCIT's req

192.168.100000000000000000 /16

FCIT = 2000

FCIT → 192.168.000010000000000000 /21 used

IBIT = 1000

192.168.000100000000000000 /21

FX = 500

IBIT → 192.168.000100000000000000 /22 used

f<sub>1</sub> = 200

192.168.000101000000000000 /22

f<sub>2</sub> = 300

192.168.000101000000000000 /22

192.168.0.0 /16

FX Break 192.168.000101000000000000 /23 used

F2 192.168.000101000000000000 /23 used



128 64 32 16 8 4 2 1

Subnet Mask FCIT

255.255.255.0

Break FCIT into 4 facilities

|        |                           |                  |      |
|--------|---------------------------|------------------|------|
| FCIT → | 192.168.00001000.00000000 | 192.168.8.0/23   | NW   |
| CS     | " 000.00000000            | 192.168.8.1/23   | 1st  |
| DS     | " 010.00000000            | 192.168.9.254/23 | Last |
| SE     | " 100.00000000            | 192.168.9.255/23 | Boad |
| IT     | " 110.00000000            |                  |      |

DS

192.168.10.0/23

192.168.10.1/23

192.168.11.254/23

192.168.11.255/23